R15

Code No: 121AH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, May/June - 2019
ENGINEERING DRAWING
(Common to CSE, MIE, PTM)

Time: 3 hours **Max Marks: 75**

Answer all five questions
All questions carry equal marks
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1. MN is vertical line of length 90 mm and represents the minor axis of an ellipse. A point P is located 90 mm from M and 50 mm from N, so that it lies on that ellipse. Draw the ellipse and find its major axis length and eccentricity. [15]

OR

2. A train is running at a speed of 40 km/hr. Construct a plane scale to read up to a km and a minute. The scale should measure up to 50 km. The RF of the scale is 1:25000. On the scale show the distance covered by the train in 39 minutes. [15]

3. The front view of a line AB 80 mm long measures 55 mm while its top view measures 70 mm. End A is in both HP and VP. Draw the projections of the line and find its inclination with the reference planes. [15]

OR

4. A regular hexagonal lamina with its edge 50 mm has its plane inclined at 45^0 to H.P and lying with one of its edges in H.P. The plane of one of its diagonals is inclined at 45^0 to XY. The corner nearest to VP is 15mm infront of it. Draw its projections. [15]

5. The hexagonal pyramid base 30 mm side and axis 60 mm long has one of its slant edges on HP such that two of its triangular faces containing the slant edge on which it rests are equally inclined to HP. The top view of the axis appears to be inclined at 45^0 to VP. Draw its projections when its base is nearer to the observer than its apex. [15]

OR

6. A pentagonal pyramid of 25mm base side and 60mm height is resting on the HP on a corner of its base in such way a that the base makes 50^0 to HP. One of the base edges is perpendicular to VP. It is cut by a section plane perpendicular to HP and inclined to VP at 30^0 such that the section plane is 6mm in front of the top end of the axis of the pyramid. Draw the sectional top view, sectional front view and the true shape of the section. [15]

7. A cylinder 50 mm in diameter and the axis 65 mm long is resting on its base with its axis perpendicular to the H.P. It is cut by a cutting plane perpendicular to V.P inclined at 45^0 to the HP and passing through a point on the axis 25mm from the top. Draw the front view, sectional top view and development of lateral surface of the bottom part of the cylinder. [15]

OR

8. A cylinder of 65 mm diameter standing on its base on HP is completely penetrated by another cylinder of 45 mm diameter with their axes intersecting at right angle. Draw the projections showing the lines of intersection, assuming that the axis of the smaller cylinder is parallel to VP. [15]

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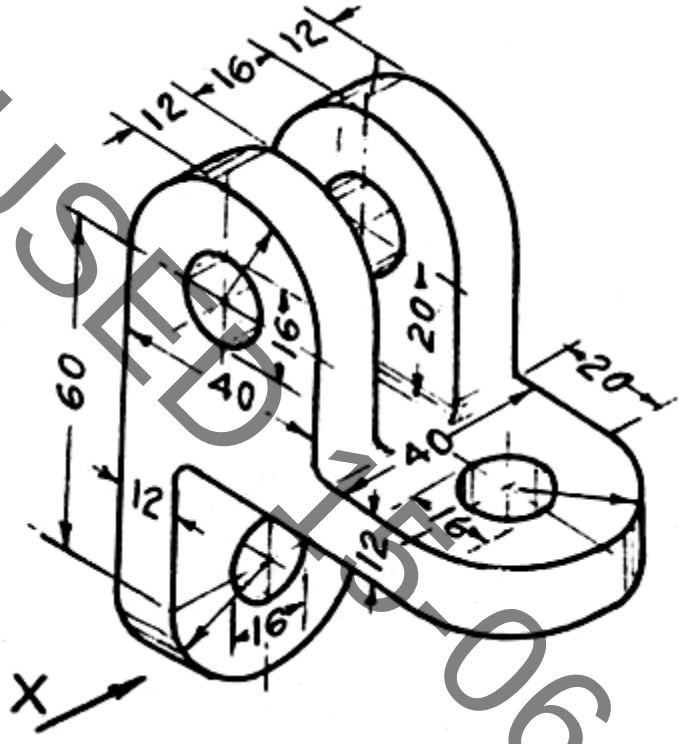
9. A right regular hexagonal prism of side of base 25 mm and altitude 55 mm has a square hole of 20 mm side at the center. The axes of the prism and the hole coincide. One of the faces of the square hole is parallel to one of the faces of the hexagon. Draw the isometric projection with the hole. [15]

OR

10. For the isometric view shown below figure, draw a) front view b) top view and c) left side view. All dimensions are in mm. [15]



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